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SARDAR PATEL UNIVERSITY
M. Sc. (IT) INTEGRATED EXAMINATION, VII SEM

SC

Date : 2nd NOV 2017, Thursday
Session : Morning
Course No : PS07CIIT01

Time : 02:00pm To 05:00pm
Sub: System Software
Total marks : 70

Q – 1 Select an appropriate option

[08]

- i) _____ part of compilation breaks up the source program into essential pieces.
a) Synthesis b) Analysis c) Execution d) None of these
- ii) _____ attempts to discover potential bugs without running the program.
a) Pretty printer b) Structure editors c) Static checker d) a & b
- iii) _____ data structure is used in synthesis phase.
a) Symbol Table b) DFD c) Name Table d) All of above
- iv) Mnemonic has _____ primary fields.
a) Mnemonic and opcode b) Name and Address
c) Mnemonic and name d) Name and opcode
- v) The _____ of a program is the address of the instruction from which its execution must begin.
a) Execution Address b) Address c) Execution start address d) All
- vi) _____ is the process of binding an external reference to the correct link time address.
a) Translation b) Linking c) Loading d) None of above
- vii) _____ is responsible for interpreting user command and implementing them by invoking different modules of the application code.
a) Dialog Manager b) Debug Manager c) Presentation manager d) All
- viii) _____ helps in obtaining information for localization of errors.
a) Debug Monitors b) Profile monitor c) Editors d) All of above

Q – 2 Answer the following Questions (Any SEVEN)

[14]

- i) What do you mean by Compiler
- ii) Define : Application Domain & Execution Domain.
- iii) Define system software and write types of software
- iv) What is Machine instructions format?
- v) What is Condition code? How it is tested?
- vi) Differentiate between linking and loading
- vii) Explain address sensitive program
- viii) Write difference between Hypertext, Hyper card and Manually.
- ix) Explain the factor on which efficiency of programs depends.

- Q – 3 a) What is language processing? Explain spectrum of language processor. [06]
b) List the phases of a compiler and explain symbol table management, error detection reporting. [06]

OR

- Q – 3 b) Give the schematic of interpretation of HLL program and execution of a machine language program by the CPU. [06]

- Q – 4 (a) Explain features provided by Assembly language programming in details [06]
(b) What is assembly language? What kind of statements are present in an assembly language program? Discuss. Also highlight the advantages of assembly language. [06]
- OR**
- Q – 4 (b) Explain design specification of an assembler. [06]
- Q – 5 a) Explain different editors in detail. [06]
b) Explain Translated, Linked and Load time address. [06]
- OR**
- Q – 5 b) Discuss various steps for execution of program written in a programming language. [06]
- Q – 6 a) Define software tools. Discuss three software tools for programming development [06]
b) Briefly discuss the principles of command dialog design. [06]
- OR**
- Q – 6 b) Explain Debug Monitor and User Interfaces in detail [06]

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